College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

ASSIGNMENT 03 — MULTIPLE CHOICE QUESTION PAPER

Sessions Covered: 05–12 and 15 (Methods, OOP Basics, Encapsulation, Constructors, Inheritance) | Excluded: 13, 14

Duration: 80 Minutes **Total Marks:** 40 **Questions:** 40 (1 mark each) **Answer all questions** on your own answer sheet.

Instructions to Candidates

1. This is a **multiple-choice paper**. Each question has **five** choices (A–E). Select the **one best** answer.
2. Answer **all** questions in the answer sheets provided by you.
3. Do **not** write answers on this question paper.
4. Write clearly using blue or black pen. Number your answers to match the question numbers.
5. Some questions are moderately difficult and may include short code fragments — read carefully.
6. Mobile phones and programmable devices are not allowed.

Section A — Methods, Paradigm & Classes (Sessions 05–09) [14 Marks]

1. In Java, a method that performs an action but does *not* return a value to the caller should use which return type?

A) int
B) String
C) void
D) boolean
E) double

2. Which statement best describes a procedural program?

A) Software is built as objects that own both data and behaviour
B) The program is mainly a sequence of steps/functions working on shared data
C) Every variable must be private
D) Inheritance is required in every program
E) Only constructors are allowed

3. The four pillars of OOP are:

A) Variables, Loops, Arrays, Methods
B) Encapsulation, Inheritance, Polymorphism, Abstraction
C) Class, Object, Package, Import
D) public, private, protected, default
E) JDK, JVM, JRE, IDE

4. In OOP, a class is best described as:

A) A running program in memory
B) A blueprint/template for creating objects
C) A single variable of type int
D) The main method only
E) A package name

5. Which line correctly creates an object of class `Rectangle`?

A) `Rectangle rect;`
B) `Rectangle rect = new Rectangle();`
C) `new = Rectangle();`
D) `Rectangle.create();`
E) `object Rectangle = rect;`

6. A field (attribute) in a Java class is:

A) A variable declared inside a method only
B) A variable declared in the class body (outside methods) that stores object data
C) Always the same as a constructor
D) A package statement
E) An import keyword

7. If `car` is an object of class `Vehicle`, which call is correct?

A) `Vehicle.accelerate();`
B) `accelerate(car);`
C) `car.accelerate();`
D) `new accelerate();`
E) `class.accelerate(car);`

8. In an OOP method, parameters are best described as:

A) Values sent into the method when it is called
B) Always private fields of the class
C) The name of the class file
D) Only used in `main`
E) Keywords reserved by Java

9. Study the method:

```
public double area(double w, double h) {
    return w * h;
}
```

What does the `return` statement do?

A) Prints `w * h` on the screen only
B) Ends the method and sends the product back to the caller
C) Creates a new object of type `double`
D) Changes the method name to `area`
E) Makes `w` and `h` into fields

10. Which method signature correctly returns a calculated area as a `double`?

A) `public void area()`
B) `public area double()`
C) `public double getArea()`
D) `double public getArea()`
E) `public getArea double`

11. Two objects are created: `Car a = new Car();` and `Car b = new Car();`. Which statement is TRUE?

A) `a` and `b` must always have identical field values forever
B) `a` and `b` are separate objects; each can store its own field values
C) Java allows only one object per class
D) `b` replaces `a` in memory automatically
E) Fields of `a` are shared with every other class

12. A variable declared *inside* a method (not in the class body) is called a:

A) field / instance variable
B) local variable
C) constructor
D) package
E) superclass

13. In OOP thinking, which best matches "an object knows its data and can do actions"?

A) All data lives only in `main` as local variables
B) Data (fields) and behaviour (methods) belong together in a class
C) Methods are forbidden; only fields are used
D) Every program must use inheritance
E) Objects cannot call methods

14. What is wrong with this call if `deposit` expects one `double` argument?

```
account.deposit();
```

A) You cannot call methods on objects
B) The call supplies no argument, so it does not match the method's parameter list
C) `deposit` must always be `void`
D) Semicolons are not allowed after method calls
E) The class name must be used instead of `account`

Section B — Encapsulation & Access (Sessions 10–11) [13 Marks]

15. Encapsulation mainly means:

A) Writing longer method names
B) Hiding an object's data and allowing access only through controlled methods
C) Using only public fields
D) Inheriting from many parent classes
E) Avoiding constructors

16. Which access modifier makes a field visible only inside the same class?

A) public
B) protected
C) private
D) open
E) global

17. Why is making a bank account `balance` field `public` a bad idea?

A) Java does not allow public fields
B) Other code can change the balance directly and break rules (e.g. set a negative balance)
C) Public fields make the program run slower always
D) The compiler deletes public fields
E) Public fields cannot store numbers

18. A getter method is used to:

A) Create a new object
B) Read/return the value of a private field safely
C) Delete a class file
D) Import a package
E) Start the JVM

19. A setter method is useful because it can:

A) Prevent the class from compiling
B) Validate data before assigning it to a private field
C) Replace the need for a class name
D) Turn private into public automatically
E) Remove inheritance

20. For a private field `title` of type `String`, the usual JavaBeans getter name is:

A) `title()`
B) `getTitle()`
C) `read_title()`
D) `TitleGet()`
E) `fetch()`

21. Study the class fragment:

```
public class BankAccount {
    private double balance;
    public double getBalance() { return balance; }
}
```

From `main` in another class, which line is LEGAL?

A) `account.balance = 100;`
B) `double x = account.getBalance();`
C) `account.balance++;`
D) `System.out.println(account.balance);`
E) `private account.balance;`

22. A setter for `balance` rejects negative amounts. What is the main benefit?

A) The program never needs a constructor
B) Invalid states are blocked at the point of change
C) All fields become public
D) Inheritance is disabled
E) The JVM runs faster only

23. Inside a class method, `this.balance` typically means:

A) A local variable named `balance` in `main` only
B) The `balance` field of the current object
C) The parent class name
D) A package import
E) A static method call

24. Which access level allows access from subclasses *and* from the same package (typical meaning of `protected`)?

A) private
B) protected
C) There is no such modifier in Java
D) void
E) final

25. Compared with public fields, private fields + public getters/setters mainly improve:

A) Control over how data is read and changed
B) The number of keywords Java has
C) Screen resolution
D) The need to delete `main`
E) Internet download speed

26. Study this setter:

```
public void setAge(int age) {
    if (age >= 0 && age <= 120) {
        this.age = age;
    }
}
```

If a caller executes `person.setAge(-5);`, what is the best description of the result?

A) `age` becomes −5
B) The invalid value is ignored; `age` is not updated to −5
C) The class is deleted
D) Java always throws an error for every setter
E) `this` becomes null

27. For a private `boolean available` field, the usual JavaBeans getter name is:

A) `getAvailable()` or `isAvailable()`
B) `availableGet()`
C) `bool()`
D) `check()` only
E) `privateAvailable()`

Section C — Constructors & Inheritance (Sessions 12, 15) [13 Marks]

28. A constructor in Java:

A) Must return `int`
B) Has the same name as the class and has no return type
C) Is called only with `System.out.println`
D) Cannot take parameters
E) Must be named `main`

29. When is a constructor executed?

A) Every time any method is called
B) When the program ends
C) When an object is created with `new`
D) Only if the field is public
E) Only inside `jshell`

30. Inheritance in Java is declared using the keyword:

A) implements
B) inherits
C) extends
D) include
E) import

31. Inheritance models which relationship?

A) has-a
B) is-a
C) uses-only
D) equals-to
E) sorted-by

32. Study the code:

```
class Vehicle { }
class Car extends Vehicle { }
```

Which statement is TRUE?

A) `Vehicle` is the child class of `Car`
B) `Car` inherits non-private members of `Vehicle`
C) A class cannot extend another class in Java
D) `extends` creates a package
E) `Car` must redefine `main`

33. In a child-class constructor, `super(...)` is used to:

A) Delete the parent class
B) Call the parent class constructor
C) Make all fields public
D) Stop inheritance
E) Print text to the console

34. If you write no constructor in a class, Java provides:

A) Nothing — the class cannot be compiled
B) A default no-argument constructor
C) A constructor that returns `int`
D) An automatic `main` method
E) Multiple inheritance

35. Which declaration is a valid constructor for class `Book`?

A) `public void Book(String t) { }`
B) `public Book(String t) { title = t; }`
C) `public int Book(String t) { return 0; }`
D) `public static Book() { }`
E) `constructor Book(String t) { }`

36. A child class cannot access a parent's `private` field directly. What is the usual correct approach?

A) Change the field to public in every program, always
B) Use public/protected getters/setters (or protected access) provided by the parent
C) Delete the parent class
D) Use `import private;`
E) Private fields are copied into `main` automatically

37. In Java, a class may extend:

A) Many classes at once with one `extends` list
B) At most one other class (single class inheritance)
C) Zero classes only — inheritance is banned
D) Only classes named `Object` written by the student
E) Only interfaces using `extends`

38. Given `class Car extends Vehicle`, which assignment is valid (is-a)?

A) `Car c = new Vehicle();` (always safe without casting)
B) `Vehicle v = new Car();`
C) `Car c = new String();`
D) `Vehicle v = new int();`
E) `extends v = new Car();`

39. In a subclass constructor, if you call `super(name)`, where should that call appear?

A) Anywhere after other statements
B) As the first statement in the subclass constructor
C) Only inside `main`
D) Only in the parent class file
E) After the object is destroyed

40. Study the code:

```
class Animal {
    public void speak() { System.out.println("..."); }
}
class Dog extends Animal {
    public void speak() { System.out.println("Woof"); }
}
```

What does `new Dog().speak();` print?

A) Nothing — child methods are ignored
B) Woof
C) A compile error always, because methods cannot share names
D) ...
E) Dog

******* END OF QUESTION PAPER *******